

Experiment No:11

Aim

To perform an **Affine Transformation** on an image using Python and OpenCV.

Procedure

1. Import the required libraries (**cv2** and **numpy**).
2. Read the input image.
3. Define three source points and three corresponding destination points.
4. Compute the affine transformation matrix using **cv2.getAffineTransform()**.
5. Apply the transformation using **cv2.warpAffine()**.
6. Display the original and transformed images.
7. Close all windows after a key is pressed.

Code

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
rows, cols = img.shape[:2]
pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])
M = cv2.getAffineTransform(pts1, pts2)
affine_img = cv2.warpAffine(img, M, (cols, rows))
cv2.imshow("Original Image", img)
cv2.imshow("Affine Transformed Image", affine_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The affine transformation was successfully performed on the input image. The transformed image showed changes in **position, rotation, scaling, and skewing** while preserving the parallelism of lines.